1.6.10

1. By modus tollens, I am not sore, so, again by modus tollens, I did not play hockey.
2. Either I worked last Monday or it was sunny on Friday. If I worked on Friday, then, by modus ponens, it was either sunny or partially sunny, so, by disjunctive syllogism, it was sunny. The claim follows by resolution.
3. Spiders are not insects, by modus tollens.
4. Homer is not a student, by modus tollens. (We can't say anything about whether Maggie is a student.)
5. We can't draw any conclusions here! We could pove that you do not eat tofu, but that's already a sentence.
6. I see elephants running down the road. By disjunctive syllogism, I am hallucinating, so, by modus tollens, I see elephants.

1.6.14

1. Universal instantiation and modus ponens.
2. Five instances of both universal instantiation and modus ponens, followed by conjunction.
3. Existential instantiation, universal instantiation, modus ponens, existential generalization.
4. See #3.

1.6.18

We cannot control which s it is such that $S(s, \text{Max})$. We can only show that there is one. Saying that it has to be Max is wrong.

1.6.28

By existential instantiation, $\neg P(c)$ for some c. Applying universal instantiation, we get $P(c) \vee Q(c)$, $\neg Q(c) \vee S(c)$, and $R(c) \to \neg S(c)$. Then we apply disjunctive syllogism to get $Q(c)$, and again to get $S(c)$, modus tollens to get $\neg R(c)$, and existential generalization.

1.7.1

Let p and q be odd. Then we can write $p = 2m + 1$ and $q = 2n + 1$ for some m, n by definition. Then $p + q = 2(m + n + 1)$, which is even.

1.7.8

Write $n = k^2$ and $n + 2 = (k + l)^2$, where l must be positive since $n + 2 > n$. Then $2 = k^2 + 2kl + l^2 - k^2 = 2kl + l^2$. As long as n is not zero, the first term is at least 2 and the second is at least 1, so this is impossible. When $n = 0$, we just check that 2 is not a perfect square.

1.7.12

Assume the contrary: that there exists a rational number r and an irrational c such that cr is rational. Write $cr = a$ for a rational a. Then $c = a/r$, since $r \neq 0$, but the quotient of two rationals is rational.

1.7.18

We prove the contrapositive. Assume that both m and n are odd. Then $m = 2p + 1$, $n = 2q + 1$ for some p, q, and $mn = (2p + 1)(2q + 1) = 2(pq + p + q) + 1$, which is odd.

1.7.26

Assume that it is possible to choose 25 days in one year such that no month has more than two. Then there must be at least $25/2 = 12.5$ months, but there are not.

1.7.28

If n is even, then $n = 2k$ for some k, so $7n + 4 = 2(7k + 2)$, which is even. Similarly, if $7n + 4$ is even, then $7n + 4 = 2p$ for some p, so $7n = 2(p - 2)$. By 1.7.18, n must be even.

1.8.4

Since $1000 = 10^3$, we just need to check that the numbers $1^3, \ldots, 9^3$ cannot be written as the sum of two cubes. We can do this by constructing a table of the values $x^3 - y^3$ for $x = 1, \ldots, 9$, $y = 1, \ldots, x$ and verifying that none of the results are perfect cubes.

1.8.12

Assume that $n = 2 \cdot 10^{500} + 15$ is a perfect square. Then $2 \cdot 10^{100} + 16 = n + 1$, which is not a perfect square by logic similar to 1.7.8. The proof is non-constructive, since it doesn't tell you which actually is a perfect square.

1.8.16

This is false: $2^{1/2} = \sqrt{2}$.

1.8.17

We need to express the fact that there exists some x such that, whenever $P(y)$ is true, $y = x$. This is stated exactly by sentence (c). We can decompose sentence (a) into $\exists x \forall y(P(y) \to x = y \land x = y \to P(y))$. Since $x = x$, we immediately see that $P(x)$ is true and that $P(y)$ implies $y = x$. Finally, we prove that (a) and (b) are equivalent. First, we prove that (a) implies (b). If there exists some x such that $P(y)$ is true (as discussed) and such that $P(y) \to y = x$, then $P(y) \land P(z) \to y = x \land z = x \to y = z$ as required. The proof that (b) implies (a) is similar.

1.8.36

The proof is very similar to the proof that $\sqrt{2}$ is irrational. We assume that $2^{1/3} = a/b$ for a fraction a/b in lowest terms. Then $a^3 = 2b^3$, so a^3 is even, so a is even, so we can write $a^3 = (2k)^3 = 8k^3$. Then $4k^3 = b^3$, so b^3 is even, so b is even, and the fraction is not in lowest terms.

2.1.1ad

$\{1, -1\}$, $\{\}$

2.1.19

Assume that there exists some x such that $x \in A$. Since $A \subseteq B$, $x \in B$, so, since $B \subseteq C$, $x \in C$. Since this is true for any arbitrary x, $A \subseteq C$.

2.1.24

Yes, you can, because $A = \bigcup P(A)$, the union of every set in the power set of A. Therefore $A = \bigcup P(A) = \bigcup P(B) = B$.

2.1.43

This is false. Let $A = \{1\}$ and $B = \{2\}$. Then $P(A \times B) = \{\{\}, \{(1,2)\}\}$, which has two elements, but $P(A)$ and $P(B)$ both also have two elements so $P(A) \times P(B)$ has four.

2.1.44

This is true. Fix $a \in A$. Then, by construction, $\{b : (a,b) \in A \times B\} = B$, and $\{c : (a,c) \in A \times C\} = C$, so, if $A \times B = A \times C$, $B = C$.

2.2.2

$A \cap B$, $A - B$, $A \cup B$, $\overline{A} \cup \overline{B}$.

2.2.15a

First, assume $x \in \overline{A \cup B}$. Then $x \notin A \cup B$, so $x \notin A$ and $x \notin B$, so $x \in \overline{A}$ and $x \in \overline{B}$, so $x \in \overline{A} \cap \overline{B}$. These implications all work the other way as well, so the reverse direction follows.

2.2.20c

Take an arbitrary $x \in (A - B) - C$. We know that $x \in A - B$ and $x \notin C$. But, if $x \in A - B$, then $x \in A$, so $x \in A - C$.

2.2.22

We will show part (a); the solution for part (b) is similar. We know that $B \subseteq A \cup B$, so we just have to prove the other inclusion. Take an arbitrary $x \in A \cup B$. Then $x \in A$ or $x \in B$. If $x \in A$, then, since $A \subseteq B$, $x \in B$, so either way $x \in B$.

2.2.36b

1. The left-hand side is the set of pairs of elements of A with elements of $B \cup C$. The right-hand side is the set of pairs of elements of A with elements of B, and with elements of A with elements of C. These are the same.

2. The proof is similar to the previous one.

2.3.6bd

- Domain: positive integers; range: $\{1, \ldots, 9\}$.
- Domain: positive integers; range: non-negative integers.

2.3.12

1. Yes
2. No ($f(1) = f(-1)$)
3. No (there is no n such that $n^3 = 2$)
4. No ($f(1) = f(2)$)

2.3.14ab

1. Yes. For any $n \in \mathbb{Z}$, $f(0, -n) = n$.
2. No. If there exist m, n such that $f(m, n) = 2$, then exercise 1.7.8 would be contradicted.

2.3.20

1. $f(n) = 2n$
2. $f(n) = \lfloor n/2 \rfloor$
3. $f(0) = 1, f(1) = 0, f(n) = n$ for all $n \neq 0, 1$
4. $f(n) = 144$